



Ethereum's Merge: Market liquidity, efficiency and volatility in the Proof of Stake Era

Bin Liu^a, Tina Prodromou^a, Sandy Suardi^a, Caihong Xu^{b,*}

^a School of Business, University of Wollongong, Australia

^b Stockholm Business School, Stockholm University, Sweden

ARTICLE INFO

JEL classification:

G12

G14

G18

Keywords:

The Merge

Proof of Stake

Proof of Work

Cryptocurrency liquidity

Ethereum

Sustainability

ABSTRACT

This study analyzes the impact of Ethereum's Merge upgrade on market liquidity, efficiency and volatility in cryptocurrency markets. The Merge, which transitioned Ethereum from Proof of Work to Proof of Stake, aimed to improve scalability, security, and energy efficiency. Findings show a significant improvement in Ether's liquidity and a slight enhancement in market efficiency post-Merge, along with reduced trading activity and intraday volatility compared to Bitcoin. These results offer insights into investor behavior, confidence, and external market influences, relevant for market participants and policymakers.

1. Introduction

Bitcoin (BTC), the first and most influential cryptocurrency, pioneered the Proof of Work (PoW) consensus mechanism to secure decentralized networks. While PoW has been widely adopted by other cryptocurrencies,¹ its high energy consumption has sparked environmental concerns.² This has led to growing interest in alternative mechanisms like Proof of Stake (PoS)³, which offers faster transaction processing, improved scalability, and robust security without the energy demands of PoW (Milunovich 2022; Nguyen et al., 2019).

On September 15, 2022, Ethereum, the platform that powers Ether (ETH), the second-largest cryptocurrency by market capitalization, transitioned from PoW to PoS in an upgrade referred to as The Merge. This shift reduced Ethereum's energy consumption by over 99 %, comparable to the electrical usage of Austria or Ireland (De Vries 2023). As countries increasingly regulate cryptocurrency power consumption, particularly PoW networks, investments in PoW are being discouraged. Thus, studying whether PoS will become the preferred consensus

mechanism is crucial, with The Merge offering a timely opportunity to explore this shift.

We hypothesize that liquidity dynamics in cryptocurrency markets are significantly influenced by validators' actions in PoS systems, where stable holdings promote higher liquidity by contributing to a more predictable token supply, attracting buyers and sellers, fostering trust, enhancing market liquidity and allowing substantial trades with minimal price impact. ETH, a leading PoS cryptocurrency, exemplifies this dynamic, with blockchain analytics firm IntoTheBlock reporting a 5.1 % rise in staked ETH in 2024, bringing the total to 28.89 % of ETH's supply, up from 23.8 % in January. According to Dune Analytics, approximately 37.79 million ETH, valued at \$84.8 billion, is staked by over one million validators, with 15.3 % locked for over three years, indicating strong investor confidence in Ethereum's long-term prospects.⁴

We analyze The Merge's impact on ETH's liquidity, efficiency, volatility, and trading activity compared to BTC. Using order book snapshot data and Difference-in-Difference analysis, we find that ETH's

* Corresponding author.

E-mail addresses: liub@uow.edu.au (B. Liu), tprodrom@uow.edu.au (T. Prodromou), ssuardi@uow.edu.au (S. Suardi), caihong.xu@sbs.su.se (C. Xu).

¹ As of November 12, 2021, PoW cryptocurrencies accounted for approximately 65% of market capitalization in cryptocurrency. Source: <https://coincodex.com/cryptocurrencies/sector/proof-of-work>

² Digiconomist (2022) reports that Bitcoin's energy consumption is par with Sweden or Ukraine in 2022.

³ Nguyen et al. (2019) suggest that PoS has received huge attention, as a solution to energy consumption of PoW, among the alternative consensus mechanisms.

⁴ Source: <https://cryptoslate.com/ethereum-staking-defies-market-trends-with-robust-growth-in-2024/>

liquidity and efficiency improved post-Merge, while volatility and trading activity declined, reflecting reduced speculation. These results suggest that PoS is more favorable than PoW, aligning with Wendl et al. (2023), who report declining investments in PoW cryptocurrencies. Our study underscores PoS's benefits in enhancing market conditions and sustainability, which are crucial for long-term viability, public trust, and regulatory compliance in both traditional finance and the cryptocurrency space.

2. Data and methodology

2.1. Data

Our dataset comprises transaction data and order book snapshots from Kaiko, focusing on Ether (ETHUSD) and Bitcoin (BTCUSD) trading pairs on Binance. The sample period spans March 1, 2022, to March 30, 2023, covering approximately six months before and after The Merge (September 15, 2022). To isolate The Merge's impact, we exclude a window preceding the event in the pre-Merge period and another following the event in the post-Merge period, accounting for anticipation effects, start-up impacts, and the FTX collapse in November 2022.⁵ Transaction data includes prices, volumes, timestamps, and trade indicators. Order book snapshots track all bids and asks within 10 % of the mid-price, updated bi-minute. Data is aggregated into 5-minute intervals and converted to daily frequency for analysis.

2.2. Liquidity measures

Building on prior research on market liquidity in cryptocurrency markets (Brauneis and Mestel, 2018; Dimpfl, 2017; Brauneis et al., 2021), we first analyze transactions and order book snapshots to measure relative quoted spread, effective spreads, and market depth. Additionally, we calculate liquidity proxies using transaction data including the Amihud (2002) illiquidity ratio, Abdi and Rinaldo (2017) estimator, Corwin and Schultz (2012) estimator and Roll (1984) measure. We also incorporate the volatility-to-volume measure in Kyle and Obizhaeva (2016). These liquidity measures are computed for each 5-minute interval over a 24-hour period and then aggregated to a daily frequency.

Quoted spread is the difference between the best ask and bid prices, divided by the quote midpoint and averaged across 5-minute intervals at time t :

$$QS_t = \frac{1}{N_t} \sum_{j=1}^{N_t} \frac{p_j^a + p_j^b}{MQ_j}, \quad (1)$$

where subscript j denotes the j th order book snapshot in interval t and MQ_j is the midpoint price. N_t is the total number of order book snapshots in interval t . We estimate effective bid-ask spread by combining order book snapshots with the first transaction that occurs after the snapshot. The price of this transaction is denoted, P^+ . The average effective spread in interval t is then calculated as:

$$ES_t = \frac{1}{N_t^+} \sum_{j=1}^{N_t^+} \frac{2 * |p_j^+ - MQ_j|}{MQ_j}, \quad (2)$$

where N_t^+ is the number of order book snapshots that a transaction follows before the next order book snapshot is recorded. Market depth is calculated as $\frac{Q_b + Q_a}{2}$, where Q_b and Q_a is the amount provided at the best bid and ask price at the time of the transaction.

The Amihud (2002) illiquidity ratio is calculated by,

⁵ Bouri et al. (2023) highlight the FTX collapse's liquidity crisis, causing Bitcoin to drop 28% and Ethereum 30% by November 9, 2022. We believe the collapse affected Ethereum and Bitcoin similarly during this period.

$$Amihud_t = \frac{1}{t} \sum_i \frac{|ret_{t,i}|}{\$Vol_{t,i}}, \quad (3)$$

where $ret_{t,i}$ denotes the return in subinterval i in t , respectively.

The Corwin and Schultz (2012) estimator is calculated from the high and low prices of two adjacent subintervals $i, i + 1$, defined as:

$$CS_{i,i+1} = [2(\exp(\alpha) - 1)] / [1 + \exp(\alpha)], \quad (4)$$

$$\text{where } \alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}}, \quad \beta = |\ln\left(\frac{H_i}{L_i}\right)|^2 + |\ln\left(\frac{H_{i+1}}{L_{i+1}}\right)|^2,$$

$\gamma = |\ln\left(\frac{H_{i+1}}{L_{i+1}}\right)|$ and H_i and L_i denote the high and low prices, respectively, in subinterval i , while H_{i+1} and L_{i+1} refer to the high and low prices, respectively, of two adjacent subintervals i and $i + 1$. The CS_t estimator for period t is the average of all CS estimators for adjacent subintervals in t .

The Abdi and Rinaldo (2017) estimator is based on the natural logarithms of high, low and closing prices (C_i) in subinterval i , denoted $h_i = \ln(H_i)$, $L_i = \ln(L_i)$ and $c_i = \ln(C_i)$, respectively and $\bar{p}_{ti} = \ln\left(\frac{h_i + L_i}{2}\right)$ the midpoint between the high and the low log prices in subinterval i . We use the two-day corrected version of the estimator which uses high and low-price data from two adjacent subintervals i and $i + 1$. It is defined as:

$$AR_i = \sqrt{\max\{4(c_i - \bar{p}_{ti})(c_i - \bar{p}_{t+1}), 0\}}, \quad (5)$$

The AR_t estimator for interval t is the average of the $AR_{t,i}$ measures for all adjacent subintervals i in t , $AR_t = \frac{1}{I-1} \sum_{i=1}^{I-1} AR_{t,i}$.

We also utilise the Roll (1984) measure. For each interval t , we obtain a spread estimate from the closing prices of all subintervals i in t . If the serial covariance is positive, we set the estimator to 0. We calculate Roll's measure based on price changes:

$$Roll_t = 2 * \sqrt{-\min\left(\text{cov}\left[\frac{\Delta C_{t,i}}{C_{t,i-1}}, \frac{\Delta C_{t,i-1}}{C_{t,i-2}}\right], 0\right)}, \quad (6)$$

where Δ is the first difference.

The Kyle and Obizhaeva (2016) derive an illiquidity index based on the ratio of volatility to dollar volume of an asset within a given interval, defined as:

$$Kyle_t = \left[\frac{\sigma_t^2(r)}{\sum_i \$volume_{t,i}} \right]^{\frac{1}{3}}, \quad (7)$$

where the volatility estimator $\sigma_t^2(r)$ is the mean of the squared returns of all subintervals i in interval t .

2.3. (In)efficiency, and volatility measures

We measure inefficiency based on variance ratios in Chordia, Roll, Subrahmanyam (2008, 2011). The inefficiency measure is calculated as:

$$InEfficiency_i = |VarianceRatio_i - 1|, \quad (8)$$

where $VarianceRatio_i = \frac{3 * Variance_{5min,i}}{Variance_{15min,i}}$ or $\frac{12 * Variance_{5min,i}}{Variance_{60min,i}}$, and the volatility measures are the squared returns during 5-min, 15-min or 60-min intervals, denoted by $Variance_{5min,i}$, $Variance_{15min,i}$, $Variance_{60min,i}$ respectively.

2.4. Trading activity measures

In order to capture the trading activities, we calculate the log number of transactions $Tnbr_t = \log(\sum_i Tnbr_i)$, log dollar volume $l\$Volume_t = \log(\sum_i Price_i * Volume_i)$, log transaction volume, $lVolume_t =$

Table 1

Summary statistics

This table presents summary statistics for the main variables of interest, as described in Section 2.2–2.4, during the pre-Merge and post-Merge periods for BTC (Control) and ETH (Treatment). ***, ** and * representing 1 % 5 % and 10 % significance level respectively.

Before The Merge: March 01, 2022 to June 30, 2022					After The Merge: December 01, 2022 to March 30, 2023				
Panel A: BTC (Control)									
Variable	Mean	Std.	Min	Max	Mean	Std.	Min	Max	Post-Pre
QS (bps)	0.2629	0.2139	0.0592	1.3659	0.3424	0.1452	0.0407	0.6482	0.0795***
ES (bps)	0.8607	0.7428	0.1415	4.4347	1.0058	0.5489	0.1542	2.9459	0.1451*
Depth	0.2927	0.0970	0.1140	0.7710	0.0845	0.0632	0.0345	0.3501	−0.2083***
Amihud (bps)	0.0010	0.0010	0.0003	0.0065	0.0027	0.0011	0.0005	0.0061	0.0017***
CS (bps)	1.0044	0.8748	0.1945	5.5146	1.0732	0.5043	0.2876	2.7316	0.0688
AR (bps)	2.3418	1.1758	0.9661	7.4684	1.3550	0.6024	0.2779	3.4581	−0.9868***
Roll (bps)	5.6686	2.8010	2.4083	17.6446	3.3718	1.6414	0.6629	9.1033	−2.2968***
Kyle (bps)	0.0002	0.0002	0.0000	0.0013	0.0001	0.0001	0.0000	0.0003	−0.0001***
Volatility 60 m (bps)	0.6209	0.9130	0.0336	6.1409	0.2638	0.3421	0.0031	1.7930	−0.3571**
Volatility 15 m (bps)	0.1327	0.1792	0.0128	1.0264	0.0544	0.0590	0.0010	0.3666	−0.0782***
Volatility 5 m (bps)	0.0355	0.0453	0.0037	0.2561	0.0139	0.0145	0.0003	0.0936	−0.0216***
InEfficiency1	0.1796	0.0916	0.0011	0.4478	0.2158	0.1118	0.0032	0.5381	0.0362**
InEfficiency2	0.3227	0.1983	0.0027	1.0834	0.4051	0.3402	0.0031	2.0983	0.0824**
Log Volume	5.2731	0.5461	4.0069	6.7862	6.6947	0.5184	5.1504	7.7321	1.4215***
Log Nr Trades	8.3279	0.3794	7.2940	9.5289	9.9104	0.5386	8.2189	10.8243	1.5826***
Log \$Volume	15.7011	0.4569	14.5783	16.9034	16.6460	0.5783	15.3833	17.8631	0.9450***
Before The Merge: March 01, 2022 to June 30, 2022					After The Merge: December 01, 2022 to March 30, 2023				
Panel B: ETH (Treatment)									
Variable	Mean	Std.	Min	Max	Mean	Std.	Min	Max	Post-Pre
QS (bps)	0.5809	0.3138	0.2867	2.0706	0.3440	0.0628	0.2388	0.5383	−0.4634***
ES (bps)	1.2524	1.0511	0.2907	6.3859	0.8178	0.3953	0.1510	2.2546	−0.4345***
Depth	3.4651	0.9620	1.9161	7.8223	7.8303	3.0386	3.0665	19.5078	4.3652***
Amihud (bps)	0.0019	0.0014	0.0007	0.0096	0.0014	0.0003	0.0008	0.0021	−0.0005***
CS (bps)	1.4103	1.1994	0.2768	7.4227	0.5466	0.3294	0.0627	1.7369	−0.8637***
AR (bps)	2.9186	1.4650	1.1509	10.0512	1.5186	0.5759	0.3547	3.3384	−1.4000***
Roll (bps)	7.1033	3.7113	2.9936	24.7136	3.9877	1.4658	1.1809	7.9409	−3.1156***
Kyle (bps)	0.0004	0.0005	0.0001	0.0028	0.0001	0.0001	0.0000	0.0005	−0.0003***
Volatility 60 m (bps)	1.0161	1.6089	0.0339	10.0611	0.3248	0.3274	0.0113	2.1624	−0.6914***
Volatility 15 m (bps)	0.4868	0.7578	0.0426	4.4256	0.1496	0.1265	0.0066	0.7462	−0.1482***
Volatility 5 m (bps)	0.0560	0.0816	0.0059	0.5043	0.0182	0.0142	0.0010	0.0939	−0.0378***
InEfficiency1	0.2080	0.0919	0.0127	0.4309	0.2079	0.1127	0.0021	0.4588	−0.0001
InEfficiency2	0.3277	0.2518	0.0048	2.0803	0.3287	0.2230	0.0003	1.0439	0.0010
Log Volume	7.6773	0.6186	6.4195	9.4431	7.3193	0.4792	6.0099	8.4715	−0.3580***
Log Nr Trades	7.9311	0.4523	6.9427	9.2685	7.7049	0.4141	6.6964	8.7141	−0.2261***
Log \$Volume	15.3954	0.4326	14.3088	16.5913	14.6199	0.5660	13.0978	15.8491	−0.7756***

$\log(\sum_t Volume_t)$.

2.5. Difference-in-difference analysis

To investigate whether The Merge event has any impacts on market liquidity and market efficiency, we apply a difference-in-difference

approach by running the following regression:

$$Y_{it} = \alpha + \beta_1 Post + \beta_2 ETH + \beta_3 Post * ETH + \varepsilon_{it}, \quad (9)$$

where Post is an event dummy set to 1 after The Merge, and 0 otherwise. ETH is a cryptocurrency dummy equals to 1 for ETH (Treatment) traded

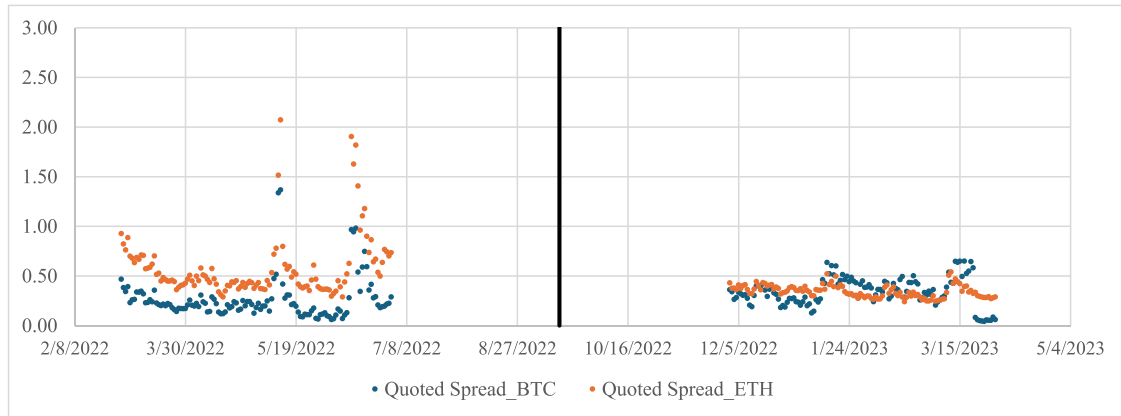


Fig. 1. Average 5-min quoted spread for BTC and ETH before and after The Merge.

Note: This figure shows average 5-minute quoted spreads for BTC (Control) and ETH (Treatment) pre- and post-Merge. The black vertical line marks The Merge on September 15, 2022. Blue (yellow) dots represent the quoted spread for BTC (ETH). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

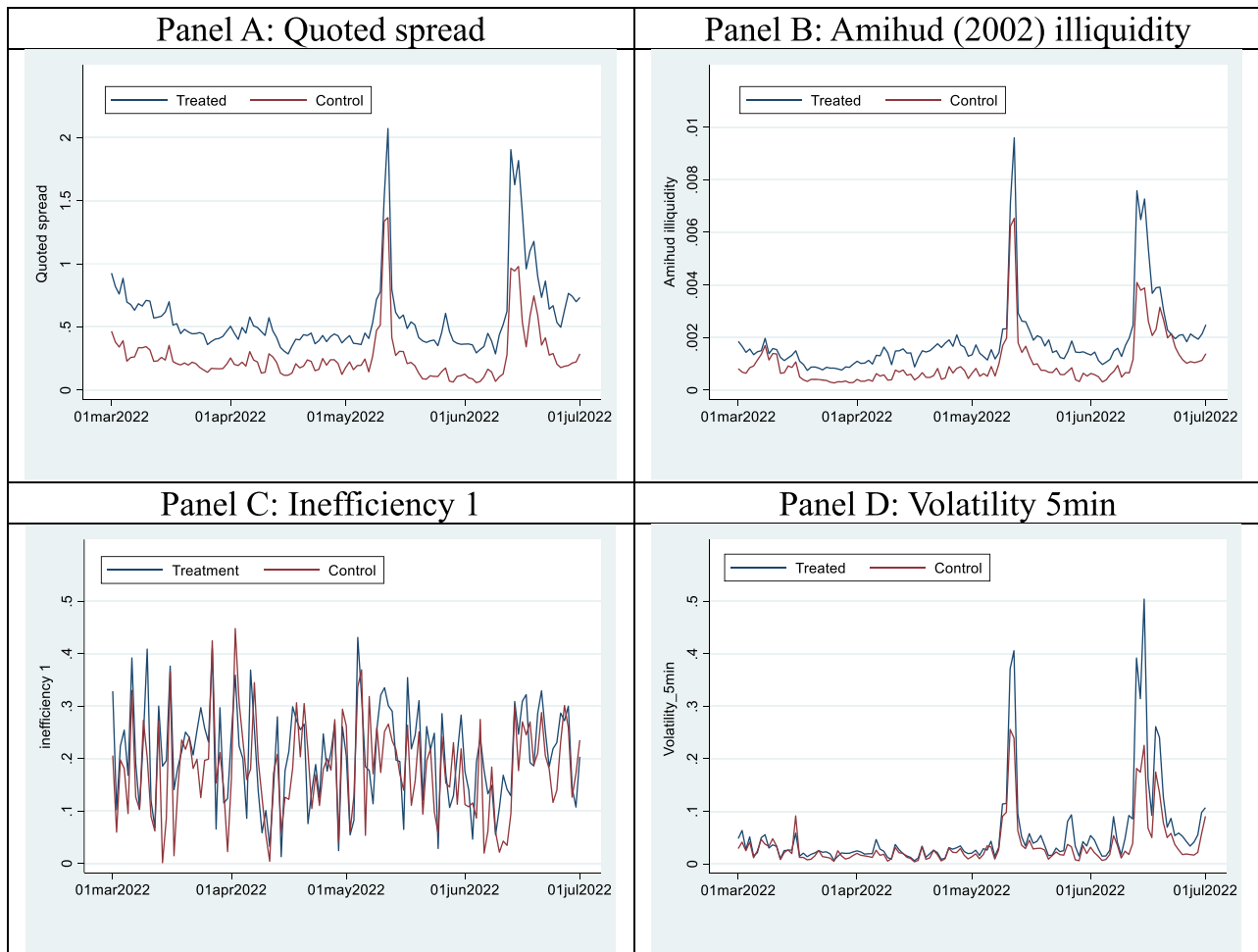


Fig. 2. Parallel trend plots for BTC and ETH before The Merge.

Note: This figure shows liquidity, inefficiency and volatility for BTC (Control) and ETH (Treatment) pre-Merge, with blue (red) line for ETH (BTC). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Difference-in-difference analysis: Liquidity impact of The Merge

This table presents the results of the difference-in-difference regression analysis of the impact of The Merge on liquidity according to $Y_{it} = \alpha + \beta_1 Post + \beta_2 ETH + \beta_3 Post * ETH + \varepsilon_{it}$. Robust s.e. are reported in parentheses. ***, ** and * representing 1 % 5 % and 10 % significance respectively.

	QS	ES	Depth	Amihud	CS	AR	Roll	Kyle
Post	0.0795*** (0.0234)	0.1451* (0.0835)	-0.2083 (0.0105)	0.0017*** (0.0001)	0.0688 (0.0912)	-0.9868*** (0.1193)	-2.2968*** (0.2934)	-0.0002*** (0.00002)
ETH	0.3180*** (0.0342)	0.3917*** (0.1161)	3.172*** (0.0872)	0.0008*** (0.0002)	0.4059*** (0.1339)	0.5768*** (0.1694)	1.4347*** (0.4193)	0.0002*** (0.00005)
Post*ETH	-0.3164*** (0.0371)	-0.5796*** (0.1313)	4.5734*** (0.2897)	-0.0022*** (0.0002)	-0.9325*** (0.1446)	-0.4132** (0.1856)	-0.8189* (0.4645)	-0.0001 (0.00005)
Obs	488	488	488	488	488	488	488	488

on Binance and 0 for BTC (Control). Y_{it} are cryptocurrency-day variables, including the quoted spread, effective spread, depth, volatility and inefficiency measures defined in sections 2.2–2.4. β_3 is the difference-in-difference estimator which captures any impact of The Merge event on Y_{it} .

3. Empirical results

3.1. Descriptive statistics and parallel trend check

Table 1 presents descriptive statistics for various measures of liquidity, volatility, market (in)efficiency, and trading activity for ETH and BTC before and after The Merge. The final column (Post-Pre)

indicates the mean difference in all measures between the post- and pre-Merge periods. Following The Merge, BTC's quoted and effective spreads increased, while ETH saw a significant decline in both spreads. Additionally, market depth decreased for BTC but increased for ETH post-Merge. Other transaction-based liquidity measures consistently show improved liquidity for ETH after The Merge, compared to BTC. Fig. 1 further illustrates these findings, revealing that BTC's quoted spread was significantly smaller than ETH's before The Merge, but both converged to similar levels afterwards.

We observe significant decreases in intraday volatility for both BTC and ETH, with ETH experiencing a notably sharper decline after The Merge. These findings may suggest a potential improvement in market efficiency for ETH relative to BTC following The Merge. Conversely,

Table 3

The Impact of The Merge on Volatility, (In)efficiency and Trading Activities

This table presents the results of the difference-in-difference regression analysis of the impact of The Merge on volatility, (in)efficiency and trading activities according to $Y_{it} = \alpha + \beta_1 Post + \beta_2 ETH + \beta_3 Post * ETH + \varepsilon_{it}$. Robust s.e. are reported in parentheses. ***, ** and * representing 1 % 5 % and 10 % significance respectively.

	Volatility 5min	Volatility 15min	Volatility 60min	Ineff1 (5min vs 15 min)	Ineff2 (5min vs 60 min)	logVolume	logNrTrade	logDollarVol
Post	−0.0216*** (0.0043)	−0.0782*** (0.0170)	−0.3571*** (0.0880)	0.0362*** (0.0131)	0.0824*** (0.0357)	1.4215*** (0.0682)	1.5826*** (0.0597)	0.9450*** (0.0668)
ETH	0.0205** (0.0084)	0.0859*** (0.0340)	0.3953** (0.1668)	0.0284** (0.1170)	0.0050 (0.0289)	2.404*** (0.0744)	−0.3968*** (0.0532)	−0.3056*** (0.0567)
Post*ETH	−0.0162* (0.0086)	−0.0699** (0.0349)	−0.3342* (0.1723)	−0.0362* (0.0186)	−0.0813* (0.0469)	−1.7795*** (0.0983)	−1.8087*** (0.0815)	−1.7205*** (0.0929)
Obs	488	488	488	488	488	488	488	488

trading activity increased for BTC while declining for ETH.

In Panel A and B of Fig. 2, we plot the liquidity variables for both BTC and ETH to assess whether Control (BTC) and Treatment (ETH) exhibit parallel trends prior to The Merge event.

As shown, ETH liquidity closely mirrors BTC indicating parallel trends. Notably, there are two significant spikes in the liquidity for both BTC and ETH in mid-May and mid-June 2022. Both BTC and ETH experienced substantial price declines, due to selloffs triggered by the collapse of Terra-LUNA (Lee et al. 2023), fears over interest rate hikes in the US, job cuts within the cryptocurrency sector, and the Celsius Network's withdrawal suspension. Similarly, in Panel C and D, we plot the inefficiency, and volatility. Again, we observe consistent trends across the main variables before The Merge event.⁶

3.2. Liquidity effect of The Merge

Table 2 presents the results of the difference-in-differences regression for the liquidity in BTC and ETH. The parameter of interest is β_3 , the coefficient for $Post * ETH$. As shown in Table 2, all eight liquidity measures indicate that liquidity in ETH improved relative to BTC, with seven of these measures being statistically significant.

We believe that liquidity in cryptocurrency markets is significantly impacted by validators' actions within the PoS consensus mechanism. Stable holdings by validators enhance liquidity by stabilizing token supply, attracting more buyers and sellers. This long-term commitment fosters trust in the network, encourages greater participation, and supports market depth, allowing for larger trades with minimal price impact.

3.3. Volatility, (In)efficiency and trading activities effects of The Merge

Table 3 presents the results of the difference-in-differences regression for volatility, (in)efficiency, and trading activities in BTC and ETH. The key parameter of interest is β_3 , the coefficient for $Post * ETH$. As indicated in Table 3, β_3 is negative for all three intraday volatility measures across different intervals, with the 15-minute interval achieving significance at the 5 % level and the other two at the 10 % level. This suggests some evidence that ETH's volatility decreased relative to BTC following The Merge.

Regarding market inefficiency, both Inefficiency1 and Inefficiency2 based on variance ratios for ETH also show a decrease relative to BTC and are significant at the 10 % level, indicating a modest improvement in market efficiency for ETH post-Merge.

However, β_3 for all three trading activity measures is negative, indicating a decrease in trading activity for ETH relative to BTC after The Merge. The transition from PoW to PoS is expected to reduce speculation in Ethereum by promoting long-term asset holding. Validators must lock their stakes, which discourages speculative behavior.

⁶ We plotted all the variables listed in Table 1, each of which demonstrates consistent parallel trends in BTC and ETH in the pre-Merge period. For brevity, we do not include all plots here.

PoS also eliminates the high energy costs of PoW mining, reducing the pressure for short-term gains. This leads to lower market volatility and fewer opportunities for rapid price speculation. Overall, these changes create a more stable market environment for Ethereum post-Merge and may attract investors who prioritize long-term holding over active trading.

4. Conclusion

In conclusion, our findings show that The Merge significantly improved ETH's liquidity relative to BTC, reduced ETH volatility, and modestly enhanced market efficiency post-transition to PoS. While ETH trading activity declined relative to BTC—likely reflecting reduced speculation and greater investor confidence—the results highlight The Merge's role in stabilizing the market. The shift from PoW to PoS fosters a more stable environment by reducing short-term trading incentives, lowering volatility, and encouraging long-term holding, enhancing Ethereum's appeal in the cryptocurrency market.

CRedit authorship contribution statement

Bin Liu: Conceptualization, Methodology, Writing – review & editing. **Tina Prodromou:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. **Sandy Suardi:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Caihong Xu:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing.

Acknowledgements

The authors acknowledge the anonymous referee for their valuable suggestions, which have enhanced the quality of this paper. Caihong is grateful to the Jan Wallander and Tom Hedelius Foundation and the Tore Browaldh Foundation for research support.

Data availability

The authors do not have permission to share data.

References

- Abdi, F., Rinaldo, A., 2017. A simple estimation of bid-ask spreads from daily close, high, and low prices. *Rev. Financial Studies* 30, 4437–4480.
- Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. *J. Financ. Mark.* 5, 31–56.
- Bouri, E., Kamal, E., Kinatder, H., 2023. FTX collapse and systemic risk spillovers from FTX token to major cryptocurrencies. *Finance Res. Lett.* 56, 104099.
- Brauneis, A., Mestel, R., 2018. Price discovery of cryptocurrencies: bitcoin and beyond. *Econ. Lett.* 165, 58–61.
- Brauneis, A., Mestel, R., Riordan, R., Theissen, E., 2021. How to measure the liquidity of cryptocurrency markets? *J. Bank. Finance* 124, 106041.
- Chordia, T., Roll, R., Subrahmanyam, A., 2008. Liquidity and market efficiency. *J. Financ. Econ.* 87, 249–268.

- Chordia, T., Roll, R., Subrahmanyam, A., 2011. Recent trends in trading activity and market quality. *J. Financ. Econ.* 101, 243–263.
- Corwin, S.A., Schultz, P., 2012. A simple way to estimate bid-ask spreads from daily high and low prices. *J. Finance* 67, 719–760.
- De Vries, A., 2023. Cryptocurrencies on the road to sustainability: ethereum paving the way for Bitcoin. *Patterns* 4 (1), 100633.
- Digiconomist 2022, 'Bitcoin historic sustainability performance—Digiconomist', <https://digiconomist.net/bitcoin-historic-sustainability-performance/>.
- Dimpfl, T. 2017, 'Bitcoin Market Microstructure'. Available at SSRN: <https://ssrn.com/abstract=2949807> or 10.2139/ssrn.2949807.
- Kyle, A.S., Obizhaeva, A.A., 2016. Market microstructure invariance: empirical hypotheses. *Econometrica* 84, 1345–1404.
- Lee, S., Lee, J., Lee, Y., 2023. Dissecting the Terra-LUNA crash: evidence from the spillover effect and information flow. *Finance Res. Lett.* 53, 103590.
- Milunovich, G., 2022. Assessing the connectedness between proof of work and proof of stake/other digital coins. *Econ. Lett.* 211, 110243.
- Nguyen, C.T., Hoang, D.T., Nguyen, D.N., Niyato, D., Nguyen, H.T., Dutkiewicz, E., 2019. Proof-of-stake consensus mechanisms for future blockchain networks: fundamentals, applications and opportunities. *IEEE access* 7, 85727–85745.
- Roll, R., 1984. A simple implicit measure of the effective bid-ask spread in an efficient market. *J. Finance* 39, 1127–1139.
- Wendl, M., Doan, M.H., Sassen, R., 2023. The environmental impact of cryptocurrencies using proof of work and proof of stake consensus algorithms: a systematic review. *J. Environ. Manage.* 326, 116530.